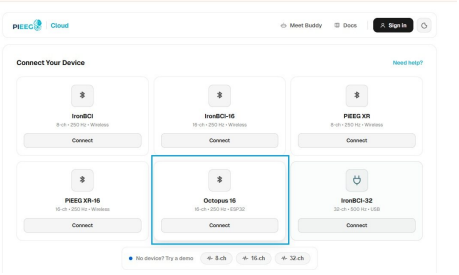
BLUETOOTH INTERFACE & SOFTWARE GUIDE · SKU PIA24027P

BLE Protocol, Packet & SDKs

How Octopus 16 streams: the BLE service, the 51-byte packet, conversion to microvolts, the software options from browser to Garmin watch, firmware source and the open datasets.



BLE 5	51 B	250/s	0.0358	16 ch
2M PHY	Packet per sample	Notifications	µV per count	In every packet

01 SOFTWARE OPTIONS

Option	Best for	Runs on	Install
PiEEG Cloud	Dashboards, spectra, maps, experiences, Buddy AI	Chrome or Edge	None — cloud.pieeg.com
octoBridge	Classrooms, block programs, OctoBio Kit	Chrome or Edge	None — octobridge.pieeg.com
octoWatch	Live wrist EEG scope and band-power HUD	Garmin watches	Connect IQ Store
JavaScript SDK	Your own web app	Chrome or Edge	`connectBLE({device:'octopus-16'})`
Python SDK	Live view, heatmap, FFT, recording	Windows, macOS, Linux	`pip install bleak`

02 BLE PROTOCOL

Advertised name	`Octopus-16-XXXX` (official firmware)
Service	`4fafc201-1fb5-459e-8fcc-c5c9c331914b`
Data (notify)	`beb5483e-36e1-4688-b7f5-ea07361b26a8`
Command (write + notify)	`c0de0001-36e1-4688-b7f5-ea07361b26a8` — register bridge to the ADCs

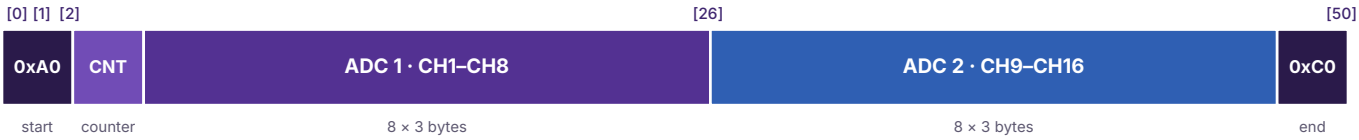


Figure 1 — One notification per sample. Channels are 24-bit big-endian two's complement. Use the counter to spot lost packets.

$\mu V = raw \times 1.2 / 4 / 8\,388\,607 \times 10^6$

1.2 V reference, gain 4: one count ≈ **0.0358 µV**.

NOTE Clock note.
Boards fitted with an 8.000 MHz clock run at ≈ 244 SPS while labelled 250. Correct for it for precise timing (ERP, P300).

PYTHON

```
raw = (b[0] << 16) | (b[1] << 8) | b[2]
if raw & 0x800000:          # negative value
    raw -= 1 << 24
uV = raw * (1.2 / 4.0 / 8388607) * 1e6
```

03 MINIMAL PYTHON READER

PYTHON · PIP INSTALL BLEAK

```
import asyncio
from bleak import BleakScanner, BleakClient

SERVICE = "4fafc201-1fb5-459e-8fcc-c5c9c331914b"
DATA    = "beb5483e-36e1-4688-b7f5-ea07361b26a8"
SCALE   = 1.2 / 4.0 / 8388607 * 1e6

def on_data(_, p):
    if len(p) != 51 or p[0] != 0xA0 or p[50] != 0xC0:
        return
    uv = []
    for c in range(16):
        b = p[2 + 3*c: 5 + 3*c]
        raw = (b[0] << 16) | (b[1] << 8) | b[2]
        if raw & 0x800000:
            raw -= 1 << 24
        uv.append(raw * SCALE)
    print(p[1], [round(v, 1) for v in uv[:4]], "...")

async def main():
    dev = await BleakScanner.find_device_by_filter(
        lambda d, adv: SERVICE in [u.lower() for u in adv.service_uuids])
    async with BleakClient(dev) as client:
        await client.start_notify(DATA, on_data)
        await asyncio.sleep(10)

asyncio.run(main())
```

04 SDKS AND TOOLS

PYTHON SDK

```
pip install bleak matplotlib numpy scipy openpyxl pandas
git clone https://github.com/pieeg-club/Octopus_16.git && cd Octopus_16
python GUI/SDK_Data_Visual.py                # 16 channels, heatmap, FFT
python GUI/SDK_Data_Visual.py --save data.csv --notch 60
python Dataset/Visualisation.py --duration 30 --out my_session.xlsx
```

JAVASCRIPT SDK

```
const eeg = new PiEEG();
eeg.onBandPowers((p) => console.log('Alpha', p.Alpha, 'Beta', p.Beta));
button.addEventListener('click', async () => {          // must be a user click
  const info = await eeg.connectBLE({ device: 'octopus-16' });
});
```

octoBridge readings: relative band power (δ θ α β γ), “focus” = β/(β+θ), “relax” = α/(α+β), RMS in µV, and “EEG ok” (RMS 2–200 µV). Ratios are uncalibrated learning indicators.

05 FIRMWARE

Item	Details
Install	octobridge.pieeg.com/flash or firmware.pieeg.com — desktop Chrome/Edge, board off the head
Source	Framework/ in the Octopus_16 repo: ADS131M08 driver, XIAO ESP32-S3 sketch (EEG over BLE, on-board FFT, Bluetooth/USB gamepad output), XIAO ESP32-C3 sketch
Libraries	NimBLE-Arduino; ESP32-BLE-Gamepad for the S3 sketch
Settings	OSR 16384 (250 SPS), gain 4; same service and 51-byte packet as the official firmware

06 OPEN DATASETS

File	Content	Size
`Blinking_Fp.xlsx`	Blinking, sensor over Fp	2,503 samples (~10 s)
`closed_eys_Pz.xlsx`	Eyes closed, sensor over Pz	2,503 samples (~10 s)
`open_yeys_Pz.xlsx`	Eyes open, sensor over Pz	2,503 samples (~10 s)

Columns: Time (s), Counter, CH1...CH16 in raw counts (× 0.0358 → µV). Each file has a Raw data and an Info sheet.

07 RELATED DOCUMENTS

OC16-DS-001 Product Datasheet	**OC16-MD-002** Mechanical & Dimensional Drawing	**OC16-PCB-003** PCB & Manufacturing Specification
OC16-PIN-004 Electrode Array & Pinout Guide	**OC16-QSG-005** Quick Start & Safety Guide	